

daten.berlin.de Searchterms



Figure 1: logo for “daten.berlin.de searchterms” dataset

This dataset contains the searchterms that users looked for on the Berlin Open Data Portal (<https://daten.berlin.de>). Terms are collected per month (starting in February 2019, when we started using our new analytics software), and ranked by how often they were searched (i.e., the number of page impressions).

Requirements

The code to extract the searchterm statistics is written in Python. It has been tested with Python 3.12.4.

The required libraries are defined in requirements.txt. In particular, these are:

- ckan_metadata_updater
- keyring
- python-dateutil
- requests

Set up a virtual environment and install the required libraries there:

```
berlin_dataportal_searchterms % python -m venv venv
(venv) berlin_dataportal_searchterms % pip install -r requirements.txt
```

You need to set two environment variables:

- MAPP_URL: The base URL of the Mapp service, most likely <https://intelligence.eu.mapp.com>.
- MAPP_USER: The username of the Mapp user that is going to run the query to retrieve the statistics.
- MAPP_PW: The password of the Mapp user that is going to run the query to retrieve the statistics.
 - Alternatively, on MacOS, you can install the `keyring` library (`pip install keyring`) and set the password there, as a key called `mapp_api` and with the user name given by `MAPP_PW`.

daten_berlin_de.searchterms.json

Download here: `daten_berlin_de.searchterms.json`

For each searchterm that was entered in a given month, the page impressions, visits, average page duration (in seconds) and exit rate (%) are listed.

The following example illustrates the structure of the data file:

```

{
  "timestamp": "2020-05-29T15:21:32+02:00",
  "source": "Webtrekk",
  "stats": {
    "site_uri": "daten.berlin.de",
    "earliest": "2019-02",
    "latest": "2020-04",
    "months": {
      "2020-04": {
        "terms": {
          "corona": {
            "impressions": 27,
            "visits": 20,
            "page_duration_avg": 36.81,
            "exit_rate": 20.0
          },
          "verkehr": {
            "impressions": 24,
            "visits": 8,
            "page_duration_avg": 38.08,
            "exit_rate": 0.0
          },
          ...
          "new york": {
            "impressions": 1,
            "visits": 1,
            "page_duration_avg": 0.0,
            "exit_rate": 0.0
          }
        },
        "removed_items": {
          "comment": "Removed 13 searchterms as potentially personal information.",
          "count": 13
        }
      },
      "2020-03": {
        ...
      },
      ...
    }
  }
}

```

Filtering Personal Information

All searchterms that potentially contain personal information are removed from the data before publishing it here.

In particular, the following categories of searchterms are removed:

- personal names
- (postal) addresses
- geographic coordinates
- personal e-mail addresses
- phone numbers
- land lots (German „Flurstück“)

Blocklist

Instances of these categories are currently not detected automatically, but rather manually via the use of a blocklist (not included in this repository), which is being extended each time the dataset is updated (i.e., every month).

Allowlist

There are exceptions where searchterms are included in the data, even though they belong to one of the exclusion categories. In particular, we allow the following kinds of searchterms:

- **Personal names of public figures**

The criterion for being a public figure is: there is a (stable) Wikipedia page for that person. The criteria for people to have Wikipedia page are defined here.

Another possible criterion is that a name has an entry in a bibliographic authority file (something like a database of all known authors), such as the Gemeinsame Normdatei. In other words, a name is the name of a published author.

- **Functional e-mail addresses**

Functional e-mail addresses (addresses not tied to a particular person, but to a role or a post such as `info@example.com`, `opendata@berlin.de` etc.) do not contain personal information and can therefore be included.

Searchterm Normalization

Searchterms are currently not normalized in any way. This means that different spellings of the same term (most importantly: differences in case) are treated as different searchterms. It is possible to sum page impressions for each term. This is not possible for visits, because the same visit might include two or more searchterms, and so the actual number of visits for a set of searchterms might be less than the sum of visits for all of them.

For example:

```

{
  ...
  "terms": {
    "corona": {
      "impressions": 27,
      "visits": 20,
      "page_duration_avg": 36.81,
      "exit_rate": 20.0
    },
    ...
    "Corona": {
      "impressions": 8,
      "visits": 6,
      "page_duration_avg": 21.25,
      "exit_rate": 0.0
    },
    ...
    "covid": {
      "impressions": 2,
      "visits": 2,
      "page_duration_avg": 13.0,
      "exit_rate": 0.0
    },
    ...
  }
}

```

People searched for **corona** (lower case), **Corona** (upper case), **covid** and possibly other related searchterms. It would be valid to say that the total number of page impressions for all spellings of **corona** is $27+8=35$, and $27+8+2=37$ for all Corona-related searches. However, the total number of visits for all spellings of **corona** is $20+6=26$ *or less* because some of these searches may have occurred within the same visit.

How to Update the Search Data with a New Month

Because adding a new month to the data involves manually editing the blocklist and allowlist, it is more complicated than just running a make target.

The following information is based on the old Ruby-based code and is no longer valid:

Here is what needs to be done:

Extract the Unfiltered Data

```
make unfiltered
```

This will extract the search data from Webtrekk Analytics up until the last day of the previous month. The output is written to `data/temp/daten_berlin_de.searchterms.unfiltered.json`.

Extract a List of Searchterms for the Previous Month

We need to manually go through all the new search terms, pick those that are problematic and then either add them to the blocklist or allowlist. To generate a simple list of terms (without hits, visits etc.) for a given month, do `make data/temp/terms_YYYY-MM.json`, e.g.:

```
make data/temp/terms_2020-06.json
```

Select Problematic Terms

Manually go through the list and extract all potentially problematic search terms. What I do is simply delete all *unproblematic* ones, leaving me with the list of problematic ones.

Update blocklist and allowlist

For each problematic term, decide if it really needs to be filtered out or if maybe it should be allowed after all. See Filtering Personal Information.

Each new addition to the blocklist simply needs to be added to the appropriate category (though the categories are just a way to structure the list for humans, they are not used otherwise).

Each new addition to the allowlist looks like this:

```
{
  ...
  "friedrich wilhelm förster": {
    "variants": [
      "friedrich wilhelm foerster",
      "friedrich wilhelm förster"
    ],
    "reference": [
      "https://de.wikipedia.org/wiki/Friedrich_Wilhelm_Foerster" ,
      "https://d-nb.info/gnd/118692038"
    ]
  }
  ...
}
```

So, either add a new variant to an existing entry or create a new one.

Remarks:

- It's possible that an entry has only one `variant`.

- The entry's key and the grouping of variants are irrelevant, just a way to structure the list for a human reader.
- Also, the **reference** is technically not necessary, but helpful as a reminder why the decision was made to include a searchterm in the allowlist.

Create Final, Filtered Data

Now that we have the updated block- and allowlist, we can filter the data and create the final data file, which goes into **data/target**.

```
make final
```

Logo

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Dataset URL: <https://daten.berlin.de/datensaetze/suchbegriffe-daten-berlin-de>

This page was generated from the github repository at https://github.com/berlinonline/berlin_dataportal_searchterms.

2026, Knud Möller, BerlinOnline GmbH

Last changed: 2026-07-30